

Lecture 8

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1 Modification of Codes

1.1 Expanding/Extending

Increasing n by adding parity check symbols, or adding columns to the generator matrix G .

1.2 Puncturing

Reducing n by dropping some parity checks, or removing columns from the generator matrix.

1.3 Lengthening

Increasing n, k by adding information symbols, $n - k$ remains the same. This is equivalent to only adding columns to the parity check matrix H .

1.4 Shortening

Decreasing n, k by dropping information symbols, $n - k$ remains the same. This is equivalent to only removing columns from the parity check matrix H .

Example: Consider Hamming code H whose parity check matrix is,

$$H(7, 4, 3) = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}$$

Now by dropping the fourth column of H, we can get

$$H'(6, 3, 3) = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

By dropping all even-weight columns of H, we get

$$H''(4, 1, 4) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Example: Shorten a $(2^m - 1, 2^m - m - 1, 3)$, Hamming code by dropping even weight columns in H .

Claim: G will have a minimum distance of 4

Proof:

3 columns of odd weight cannot add to $\mathbf{0}$, therefore no set of 3 columns can be linearly dependent.

Now, choose 3 columns h_j, h_k, h_l and choose 4^{th} column to be $h_j + h_k + h_l$ (odd weight column). Thus, there exist 4 linearly dependent columns, and the minimum distance $d = 4$

1.5 Augmenting

Increase k , with n unchanged. Obtained by adding rows (codewords) to G .

1.6 Expurgating

Decrease k , with n unchanged. Obtained by removing rows (codewords) from G .

Example: Expurgating the Hamming code

$$H' = \begin{bmatrix} H & \\ 1 & \dots & 1 \end{bmatrix}$$

By adding all-ones row to H , all odd weight codewords are expurgated and we obtain a $(2^m - 1, 2^m - m - 2, 4)$ code.

2 Reed-Muller Codes

Defined by the pair (m, r) , $0 \leq r \leq m$. For an r^{th} order Reed-Muller code $RM(r, m)$, $n = 2^m$, $k = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}$, $d = 2^{m-r}$

Example:

$m = 5, r = 2 \implies n = 32, k = 1 + \binom{5}{1} + \binom{5}{2} = 16, d = 8 \equiv (32, 16, 8)$ -code

$m = 5, r = 3 \implies n = 32, k = 1 + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} = 26, d = 4 \equiv (32, 26, 4)$ -code. This has the same minimum distance, but a higher rate than the Extended Hamming code $(8, 4, 4)$.

Note: $r = 0$ corresponds to a repetition code.

2.1 Generator Matrix of RM(r, m)

$$G = \begin{bmatrix} G_0 \\ G_1 \\ G_2 \\ \vdots \\ G_r \end{bmatrix}$$

- $G_0 \longrightarrow$ All 1's vectors of length $2^m : (1 \times 2^m)$
- $G_1 \longrightarrow$ All binary tuples of length m as columns in $G_1 : \left(\binom{m}{1} \times 2^m\right)$

- $G_2 \longrightarrow$ Take all pairs of rows from G_1 and include their product in $G_2 : \left(\binom{m}{2} \times 2^m\right)$
- $G_l \longrightarrow$ Take all possible l rows from G_1 and include their product in $G_l : \left(\binom{m}{l} \times 2^m\right)$

Example:

$(m = 4, r = 2) \implies n = 2^m = 16, k = 11, d = 4 \implies (16, 11, 4)$ Code

$$\mathbf{G} = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_1 \cdot a_2 \\ a_1 \cdot a_3 \\ a_1 \cdot a_4 \\ a_2 \cdot a_3 \\ a_2 \cdot a_4 \\ a_3 \cdot a_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$G_0 = [a_0] : \binom{m}{0} \text{ vectors with weight} = 2^{m-0} = 16$$

$$G_1 = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} : \binom{m}{1} \text{ vectors with weight} = 2^{m-1} = 8$$

$$G_2 = \begin{bmatrix} a_1 \cdot a_2 \\ a_1 \cdot a_3 \\ a_1 \cdot a_4 \\ a_2 \cdot a_3 \\ a_2 \cdot a_4 \\ a_3 \cdot a_4 \end{bmatrix} : \binom{m}{2} \text{ vectors with weight} = 2^{m-2} = 4$$

2.2 Recursive Construction of Generator Matrix of RM(r,m)

Constructing codeword of length 2^m for $RM(r, m)$ from codewords of length 2^{m-1} belonging to $RM(r, m-1)$ and $RM(r-1, m-1)$.

$$RM(r, m) = \{(u, u + v) : u \in RM(r, m-1), v \in RM(r-1, m-1)\}$$

$$G(r, m) = \begin{bmatrix} G(r, m-1) & G(r, m-1) \\ 0 & G(r-1, m-1) \end{bmatrix}$$

Note: $(u, u + v)$ is one-to-one map $\implies$ Dimension will add up

2.3 Parameters of Code

The dimension of the code can be shown through induction. If we assume the hypothesis to be true for $RM(i, m-1)$ $0 \leq i \leq m-1$,

$$\begin{aligned} \dim(RM(r, m)) &= \dim(RM(r, m-1)) + \dim(RM(r-1, m-1)) \\ &= 1 + \binom{m-1}{1} + \cdots + \binom{m-1}{r} + \binom{m-1}{0} + \binom{m-1}{1} + \cdots + \binom{m-1}{r-1} \\ &= 1 + \binom{m}{1} + \binom{m}{2} + \cdots + \binom{m}{r} \end{aligned}$$

For the minimum distance, assume that the hypothesis is true for $RM(i, m)$ $i \leq r$. Consider codewords $f, f' \in RM(r, m)$, $g, g' \in RM(r-1, m)$. Take $c_1 = (f, f+g)$, $c_2 = (f', f'+g')$ where $c_1, c_2 \in RM(r, m+1)$.

- If $g = g'$, $d(c_1, c_2) = 2d(f, f') \geq 2 \cdot 2^{m-r} = 2^{m+1-r}$
- If $g \neq g'$,

$$\begin{aligned} d(c_1, c_2) &= d(f, f') + d(f+g, f'+g') \\ &= d(f, f') + wt(g-g' + f-f') \\ &\geq wt(f-f') + wt(g-g') + wt(f-f') = wt(g-g') \\ &\geq 2^{m+1-r} \end{aligned}$$

Here, we have used the identity that $wt(x+y) \geq wt(x) - wt(y)$.